Impact of Generative AI on Academic Performance and Burnout

A Data-Driven Analysis of Academic Outcomes and Student Well-being

TEAM NUMBER: 10

TEAM MEMBERS:

- SHREYA SRINIVASAN (IIMSI_EDGE_2511198)
- KUMKUM BATHRI (IIMSI_EDGE_2511038)
- AKASH RAJ NIGAM (IIMSI_EDGE_2511002)
- EKLAVYA SINGH (IIMSI_EDGE_2511102)
- ANUSHI PANDEY (IIMSI_EDGE_2511064)
- KAUSTUBH GIRIDHAR (IIMSI_EDGE_2511182)

Table Of Contents

Contents	Page No.
Executive Summary	2
Business Problem	3
Research Objectives & Project Theme	4
Dataset Description	5
Dataset Overview	5
Dataset Source	5
Dataset Characteristics	5
Methodology	8
Research Methodology	8
Data Preprocessing	9
Exploratory Data Analysis	10
Feature Engineering	11
Model Development	12
Model Evaluation	12
Analysis and Findings	13
Multiple Linear Regression	13
Binary Logistic Regression	24
Business Recommendations	34
Conclusion	36
References	37
Appendix	38

Executive Summary

Generative Artificial Intelligence (GenAI) is rapidly transforming the way university students learn, research, and complete academic tasks. While AI offers significant opportunities to enhance learning efficiency and academic performance, its increasing use has also raised concerns regarding excessive dependency, reduced independent learning, and student burnout. This study investigates the academic and behavioural impact of Generative AI on university students by examining the factors influencing academic performance and severe burnout.

A structured data analytics approach was adopted using a dataset of university students. Following data preprocessing, exploratory data analysis, and feature engineering, **Multiple Linear Regression** was used to identify factors associated with students' **Post-Semester GPA**, while **Binary Logistic Regression** was employed to predict the likelihood of **severe burnout**. Model performance was evaluated using appropriate statistical and predictive evaluation metrics.

The findings indicate that **effective AI usage, particularly strong Prompt Engineering skills and learning-oriented AI applications, is positively associated with academic performance**, whereas **greater AI dependency, intensive AI engagement, examination anxiety, and increasing study workload are associated with a higher likelihood of severe burnout**. The study further highlights that traditional study habits continue to play an important role in supporting academic success, suggesting that AI should complement rather than replace independent learning.

The findings provide practical insights for universities and educators to develop evidence-based strategies that promote responsible AI adoption, strengthen AI literacy, redesign assessments for higher-order learning, and implement early intervention mechanisms to support student well-being. By integrating predictive and inferential analytics, this study contributes to a balanced understanding of how Generative AI can be effectively leveraged to enhance learning outcomes while mitigating potential risks in higher education.

Business Problem

The rapid adoption of Generative Artificial Intelligence (GenAI) in higher education has transformed how students learn, complete assignments, and prepare for examinations. While GenAI has the potential to enhance learning and academic performance, concerns have emerged regarding excessive dependence, changing study habits, and student well-being. Consequently, educational institutions require empirical evidence to understand how AI usage influences both academic outcomes and burnout among students.

This study investigates the impact of Generative AI from two complementary perspectives.

Objective 1: University Perspective

Research Question:
How do AI usage patterns, study habits, and academic characteristics influence students' Post-Semester GPA?

A **Multiple Linear Regression** model is employed to examine the relationship between students' Post-Semester GPA and factors such as AI usage, study habits, prior academic performance, and learning characteristics. The objective is to identify the key determinants of academic success and evaluate the role of AI-assisted learning in higher education.

Objective 2: Student Perspective

Research Question:
Which behavioural, academic, and AI-related factors are significantly associated with high burnout risk among students?

A **Binary Logistic Regression** model is developed to assess how students' academic characteristics, AI usage patterns, study habits, and anxiety levels are associated with the likelihood of experiencing high burnout. The objective is to identify the factors most strongly linked to burnout and provide insights that can support interventions aimed at improving student well-being while promoting responsible AI use

Research Objectives

- Develop a Multiple Linear Regression model to examine how AI usage patterns, study habits, and academic characteristics influence students' Post-Semester GPA.
- Develop a Binary Logistic Regression model to identify the factors associated with severe student burnout.
- Compare the impact of AI usage on students' academic performance and psychological well-being by analysing both academic outcomes and burnout risk.
- Recommend strategies for educational institutions to promote responsible AI usage while supporting both academic success and student well-being.

Project Theme

This study examines the dual impact of Generative AI in higher education from both the university's and the student's perspectives. It investigates how AI usage patterns, study habits, and academic characteristics relate to students' academic performance while also examining the factors associated with different levels of burnout. By combining these two perspectives, the study provides a comprehensive understanding of how AI influences both academic achievement and student well-being, enabling educational institutions to develop informed strategies for the responsible integration of AI into learning environments.

Dataset Description

2.1 Dataset Overview

The study uses the **AI Impact on Students** dataset obtained from **Kaggle**. The dataset contains information on students' academic performance, AI usage patterns, study habits, prompt engineering skills, institutional policies, and burnout risk. It was selected because it captures both academic and behavioural characteristics, making it suitable for analysing the impact of Generative AI on students from multiple perspectives.

The dataset consists of **50,000 student records**, with each record representing an individual student's learning behaviour, AI usage, academic performance, and burnout level. After preprocessing and feature engineering, the dataset was used to develop two analytical models addressing the objectives of this study.

2.2 Dataset Source

- **Source:** Kaggle
- **Dataset Name:** AI Impact on Students
- **Author:** Lavesh Jadon

2.3 Dataset Characteristics

- Dataset Overview

The dataset contains information regarding student academic performance, AI usage, and well-being.

Property	Description
Population	University students
Rows	50,000

Columns	16
Missing Values	None

- Column Descriptions

Here is the complete description of all 16 columns combined into a single table:

Column	**Type**	**Description**
Student_ID	Integer	Unique numeric identifier for each student (100001–150000)
Major_Category	Categorical	Student's field of study — STEM, Business, Humanities, Medical, Arts
Year_of_Study	Categorical	Academic standing — Freshman, Sophomore, Junior, Senior, Graduate
Pre_Semester_GPA	Float	GPA at the start of the semester (range: ~1.18 – 4.00)
Post_Semester_GPA	Float	GPA at the end of the semester (range: ~1.00 – 4.00) — **primary outcome variable**
Weekly_GenAI_Hours	Float	Average hours per week spent using Generative AI tools (0 – 40 hrs)
Primary_Use_Case	Categorical	Main purpose for which the student uses AI —

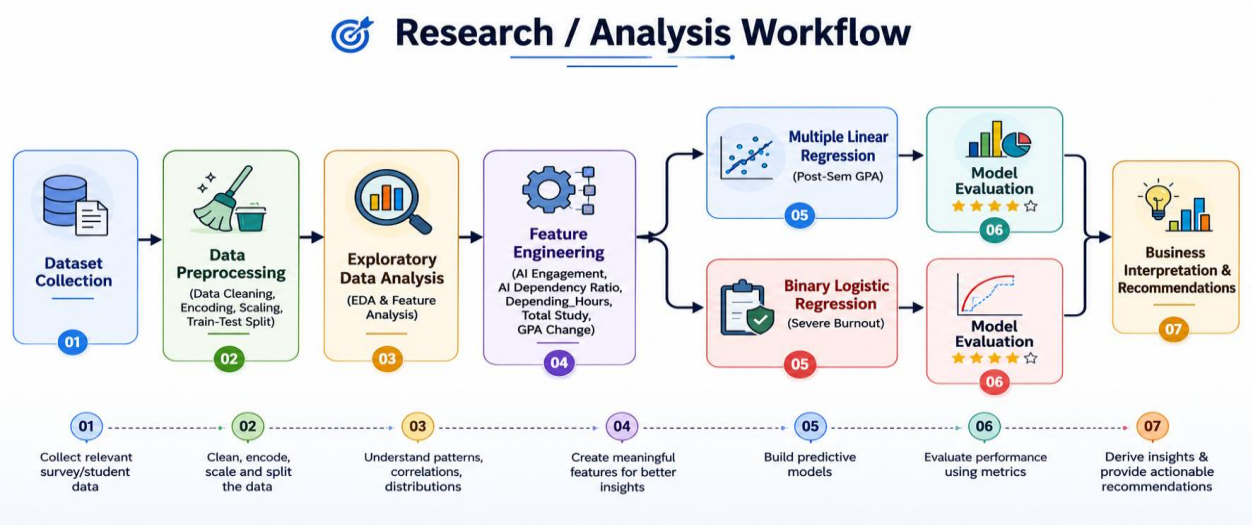
		Copywriting/Drafting, Summarizing_Reading, Debugging/Troubleshooting, Ideation, Direct_Answer_Generation
Prompt_Engineering_Skill	Categorical	Self-assessed proficiency in crafting effective prompts — Beginner, Intermediate, Advanced
Tool_Diversity	Integer	Number of distinct AI tools used (1–5)
Paid_Subscription	Boolean	Whether the student has a paid AI subscription (True / False)
Traditional_Study_Hours	Float	Weekly hours devoted to traditional (non-AI) study methods (1 – ~36 hrs)
Perceived_AI_Dependency	Integer	Student's self-rated dependency on AI tools, on a scale of 1 (low) to 10 (high)
Institutional_Policy	Categorical	The institution's official stance on AI usage — Allowed_With_Citation, Strictly_Ban, Actively_Encouraged
Anxiety_Level_During_Exams	Integer	Self-reported exam anxiety score from 1 (minimal) to 10 (severe)
Skill_Retention_Score	Float	Score (0–100) measuring how well students retained learned skills post-semester
Burnout_Risk_Level	Categorical	Assessed burnout risk — Low, Medium, High

Methodology

3.1 Research Methodology

This study follows a structured data analytics framework to examine the impact of Generative Artificial Intelligence (GenAI) on students' academic performance and well-being. The analytical process consists of data collection, preprocessing, exploratory data analysis, feature engineering, model development, model evaluation, and interpretation of findings. Two statistical models were developed to address the research objectives: a **Multiple Linear Regression** model to analyse the factors influencing students' Post-Semester GPA and a **Binary Logistic Regression** model to identify the determinants of severe student burnout.

Figure 3.1 Research Methodology Framework



3.2 Data Preprocessing

3.2.1 Data Inspection

The dataset was imported into Python using the Pandas library. Initial inspection was performed using info(), describe(), and isnull().sum() to examine the dataset structure, variable types, descriptive statistics, and missing values. The dataset contained no missing observations.

3.2.2 Data Cleaning

The **Student_ID** attribute was removed because it served only as a unique identifier and had no analytical significance. Outlier analysis was conducted using boxplots, and observations with **Traditional_Study_Hours greater than 32** were removed to eliminate unrealistic values.

3.2.3 Categorical Variable Encoding

Nominal variables (**Major_Category, Primary_Use_Case, and Institutional_Policy**) were encoded using One-Hot Encoding. Boolean variables were converted into binary values, while ordinal variables (**Year_of_Study, Prompt_Engineering_Skill, and Burnout_Risk_Level**) were encoded according to their natural ordering. One dummy variable from each nominal feature was excluded to avoid multicollinearity.

3.2.4 Feature Scaling

Continuous predictor variables were standardised using **StandardScaler** to ensure that all variables were measured on a comparable scale prior to model development.

3.2.5 Train-Test Split

The processed dataset was divided into **60% training** and **40% testing** subsets. Stratified sampling was employed for the classification model to preserve the original class distribution.

3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the dataset, identify variable relationships, and obtain preliminary insights before model development.

The analysis included:

- Dataset inspection and descriptive statistics
- Distribution analysis using histograms and Kernel Density Estimation (KDE)
- Outlier detection using boxplots
- Frequency analysis of categorical variables using count plots
- Pearson and Spearman correlation analysis
- Scatter plots and boxplots to examine relationships between predictors and Post-Semester GPA
- KDE plots for analysing predictor distributions across burnout categories
- Variance Inflation Factor (VIF) analysis to assess multicollinearity

The insights obtained from EDA guided the feature engineering process and informed the selection of variables for subsequent modelling.

3.4 Feature Engineering

To better represent students' learning behaviour and AI usage patterns, several derived variables were created from the original dataset. These engineered features capture relationships that cannot be adequately represented by the original variables alone and provide additional explanatory power for the regression models.

Engineered Feature	Formula / Definition	Purpose
Depending_Hours	Weekly_GenAI_Hours × Perceived_AI_Dependency (mean-centred)	Captures the interaction between AI usage and perceived AI dependency.
AI_Engagement	Weekly_GenAI_Hours × Tool_Diversity × Prompt_Engineering_Skill	Measures the overall intensity and effectiveness of AI usage.
AI_Dependency_Ratio	Weekly_GenAI_Hours ÷ (Traditional_Study_Hours + 1)	Represents students' reliance on AI relative to traditional study methods.
Total_Study	Weekly_GenAI_Hours + Traditional_Study_Hours	Represents the student's overall study effort.
GPA_Change	Post_Semester_GPA − Pre_Semester_GPA	Measures change in academic performance during the semester.
Is_Burn_Out	High = 1, Low/Medium = 0	Binary target variable created to identify severe burnout.

These engineered variables were incorporated into the regression models to better capture behavioural patterns associated with academic performance and severe burnout. They enabled the models to evaluate not only individual variables but also the combined influence of AI usage, study behaviour, and student learning characteristics.

3.5 Model Development

To address the research objectives, two statistical models were developed.

Model 1: Multiple Linear Regression

A Multiple Linear Regression model was developed to examine how academic, behavioural, and AI-related factors influence students' **Post-Semester GPA**. The model estimates the contribution of each predictor while controlling for the effects of all remaining variables.

Model 2: Binary Logistic Regression

A Binary Logistic Regression model was developed to identify the academic, behavioural, AI-related, and psychological factors associated with **severe student burnout**. The dependent variable (**Is_Burn_Out**) was created by combining the original Low and Medium burnout categories into a single class while retaining High burnout as the positive class. The model estimates the probability of students experiencing severe burnout based on the selected predictors.

3.6 Model Evaluation

Multiple Linear Regression

The model was evaluated using:

- Coefficient of Determination (R^2)
- Adjusted R^2
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)
- F-statistic
- Residual Analysis
- Statistical Significance (p-values)

Binary Logistic Regression

The burnout model was evaluated using:

- Classification Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Receiver Operating Characteristic (ROC) Curve and AUC
- Coefficient Analysis
- Statistical Significance (p-values)

Analysis and Findings

Model 1: Linear Regression

Objective

To develop a predictive model for Post-Semester GPA that helps the university understand how AI-assisted learning and study behaviors influence academic performance, supporting evidence-based educational policies and interventions.

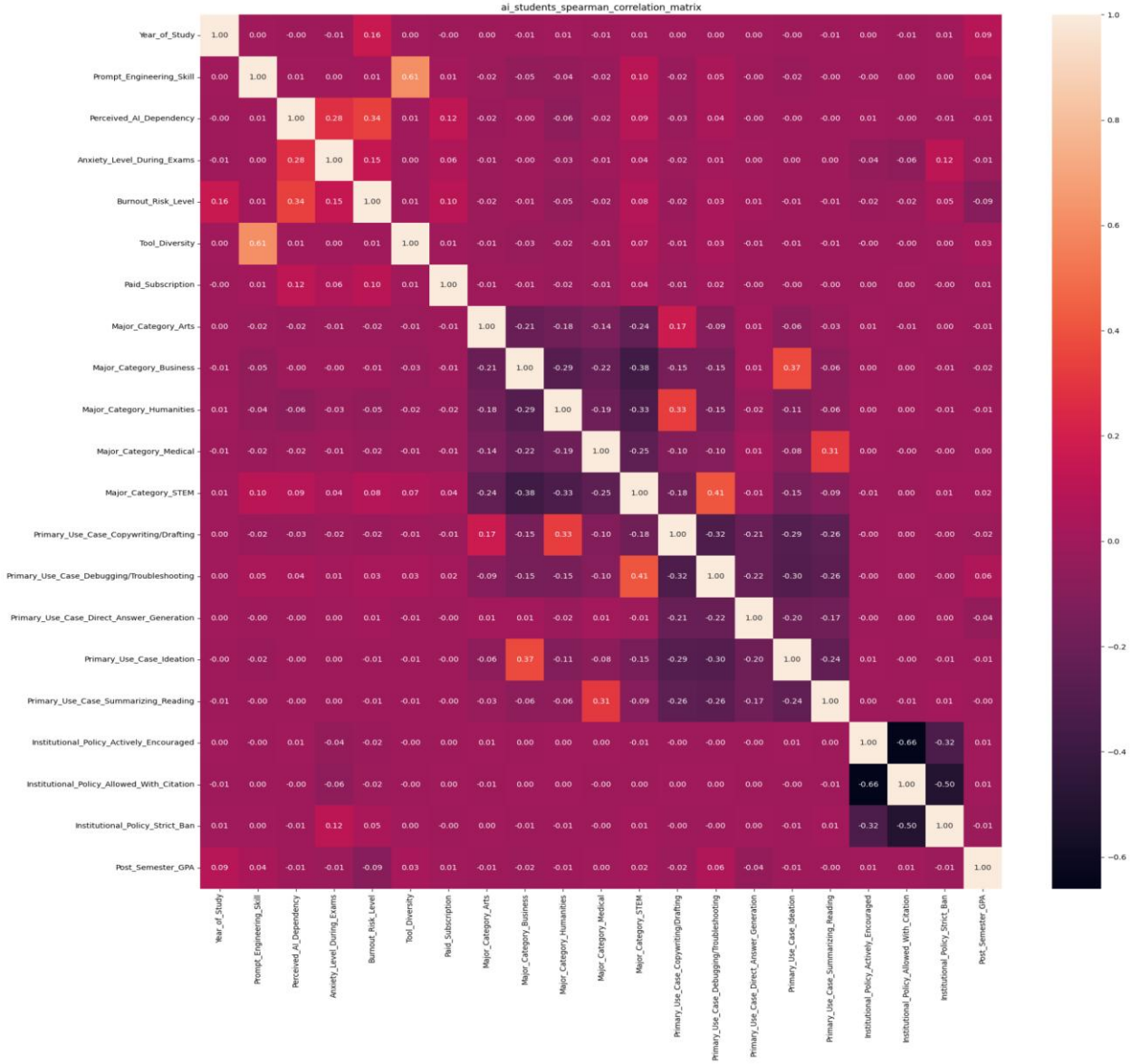
Dependent Variable

- Post_Semester_GPA

Independent Variables

- Year_of_Study
- Pre_Semester_GPA
- Weekly_GenAI_Hours
- Prompt_Engineering_Skill
- Traditional_Study_Hours
- Perceived_AI_Dependency
- Anxiety_Level_During_Exams
- Primary_Use_Case_Debugging/Troubleshooting
- Primary_Use_Case_Direct_Answer_Generation
- Depending_Hours *(Interaction between Weekly_GenAI_Hours and Perceived_AI_Dependency)*

Correlation heatmap



Interpretation

The Spearman correlation analysis of one-hot encoded variables indicates that most predictor variables exhibit **weak correlations with Post-Semester GPA**, suggesting that no single ordinal or categorical variable alone strongly determines academic performance. This supports the use of a multiple linear regression model, where the combined influence of several variables is evaluated.

Among the independent variables, **Prompt Engineering Skill** and **Tool Diversity** show a moderate positive correlation *(ρ = 0.61)*, indicating that students with stronger prompting skills tend to use a wider variety of AI tools. Similarly, **Perceived AI Dependency** is moderately associated with **Burnout Risk Level** *(ρ = 0.34)*, suggesting that students who perceive themselves as more dependent on AI are more likely to report higher burnout levels.

The matrix also reveals a positive relationship between **Anxiety Level During Exams** and **Perceived AI Dependency** *(ρ = 0.28)*, indicating that students experiencing greater examination anxiety tend to report slightly higher reliance on AI tools.

Most of the remaining correlations are relatively small ($|\rho| < 0.30$), indicating limited overlap among the predictor variables. This suggests that the selected variables contribute distinct information to the regression model and that severe multicollinearity is unlikely to be a concern.

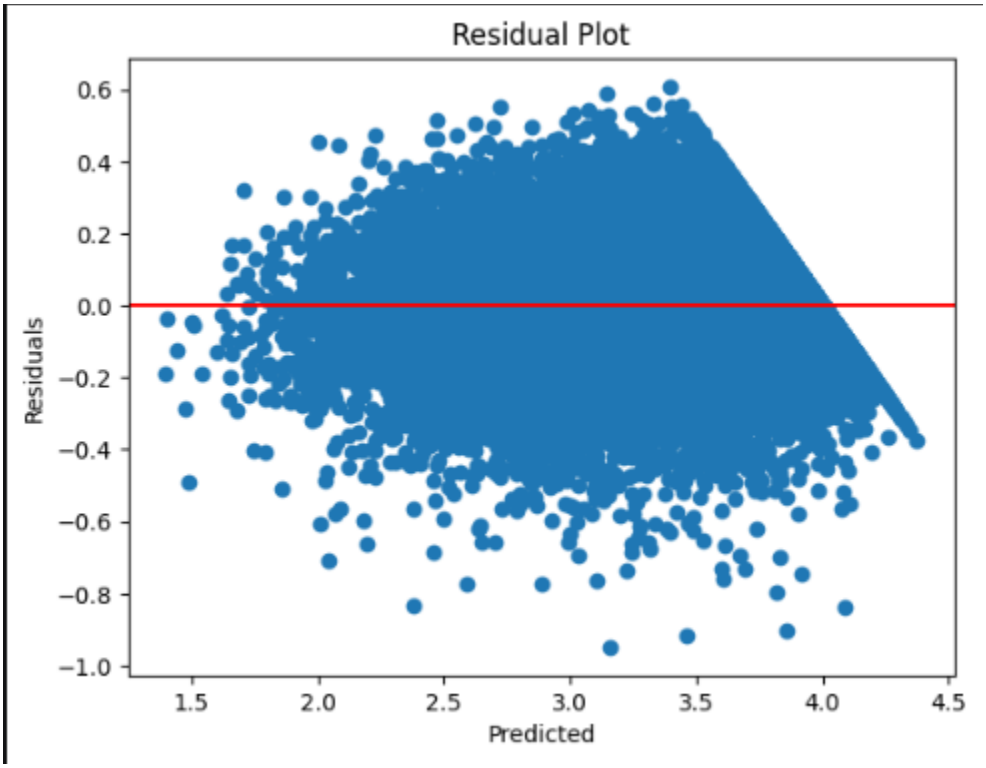
Overall, the Spearman correlation analysis provided an initial understanding of the relationships among the ordinal and categorical predictors before fitting the multiple linear regression model.

Model Assumptions & Limitations

While the model demonstrates strong predictive performance, a few limitations should be considered when interpreting the results:

- **Association, not causation:** The model identifies statistical relationships between variables and Post-Semester GPA but cannot confirm that one variable directly causes changes in another.
- **Self-reported data:** Several variables, such as AI dependency, prompt engineering skill, and study hours, were collected through self-reported responses. These may be influenced by recall errors or personal bias.
- **Linear relationship assumption:** The OLS regression model assumes a linear relationship between the predictors and GPA. In reality, the impact of AI usage and learning behaviour may be more complex and nonlinear.

- **Limited generalizability:** The findings are based on the selected sample of students. Therefore, the results may not be directly applicable to students from other universities, disciplines, or educational systems without further validation.
- **Unobserved factors:** Factors such as teaching quality, motivation, socioeconomic background, learning environment, and personal circumstances were not included in the model and may also influence academic performance.
- **Heteroscedasticity in residuals:** The residuals exhibited non-constant variance. Although HC3 robust standard errors were employed to produce reliable coefficient estimates and significance tests, the model may still exhibit varying prediction uncertainty across different groups of students.



These limitations do not undermine the usefulness of the model but highlight the need to interpret the findings within the context of the available data and a

Model Evaluation

A. OLS Regression Statistics

Metric	**Value**
Model	Ordinary Least Squares (OLS)
Number of Observations	29,999
R^2	**0.895**
Adjusted R^2	**0.895**
F-statistic	**26,15e+04**
Prob (F-statistic)	**< 0.001**
Covariance Type	HC3 (Robust Standard Errors)

Interpretation

- The model explains approximately **89.5% of the variation** in students' Post-Semester GPA.
- The **Adjusted R^2** is equal to the R^2 value, indicating that the predictors contribute meaningfully to the model without unnecessary complexity.
- The **F-statistic** with **p < 0.001** confirms that the regression model is statistically significant overall.
- Since heteroscedasticity was detected during model diagnostics, **HC3 robust standard errors** were used to produce reliable statistical inference.

B. Prediction Performance

1. Regression Coefficient Table

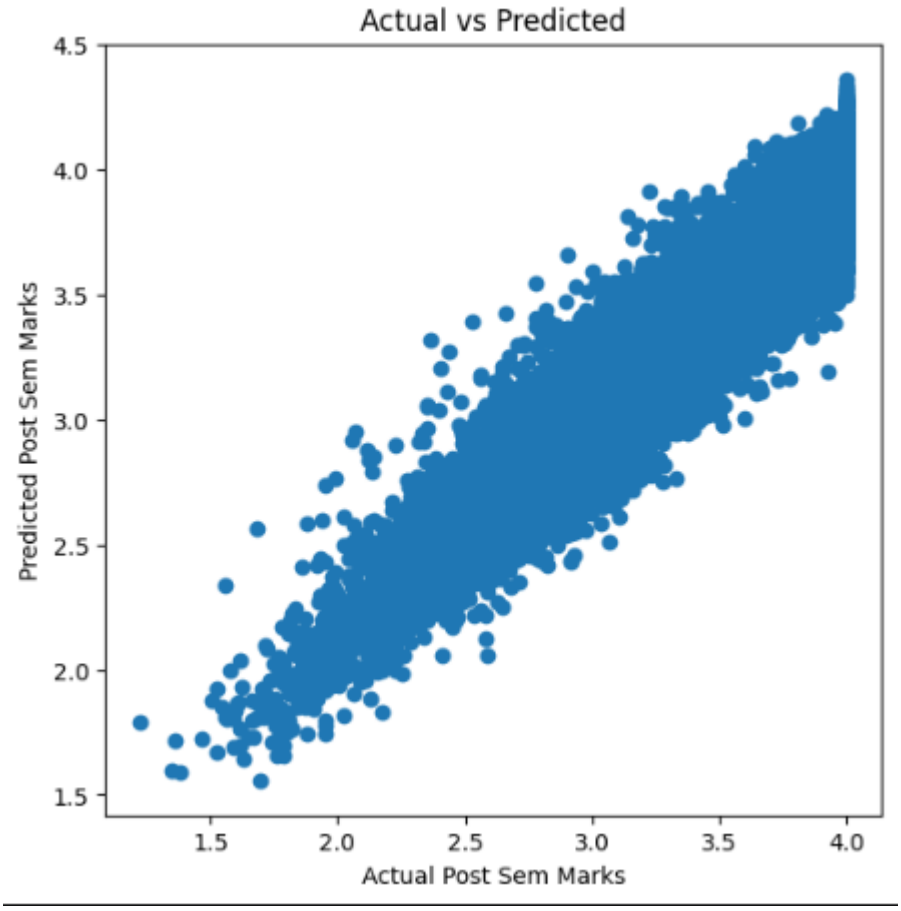
Predictors	**Coefficient (β)**	**Std. Error**	**z-value**	**p-value**
Constant	3.3493	0.001	3599.665	<0.001
Year_of_Study	0.0440	0.001	47.383	<0.001
Pre_Semester_GPA	0.4607	0.001	501.430	<0.001
Weekly_GenAI_Hours	0.0161	0.002	9.943	<0.001
Prompt_Engineering_Skill	0.0230	0.001	24.308	<0.001
Traditional_Study_Hours	0.0708	0.001	74.504	<0.001
Perceived_AI_Dependency	-0.0011	0.001	-0.908	**0.364**
Anxiety_Level_During_Exams	-0.0057	0.001	-5.696	<0.001

Primary_Use_Case_Debugging/Troubleshooting	0.0205	0.001	21.496	<0.001
Primary_Use_Case_Direct_Answer_Generation	-0.0226	0.001	-19.492	<0.001
Depending_Hours (Interaction)	-0.0194	0.002	-11.533	<0.001

Table 2. Diagnostic Statistics

Diagnostic Test	**Value**	**Interpretation**
Omnibus	428.573	Residuals deviate from perfect normality
Prob (Omnibus)	<0.001	Residuals are not normally distributed
Jarque–Bera	599.299	Supports non-normal residuals
Prob (JB)	7.31×10^{-131}	Statistically significant departure from normality
Skewness	-0.183	Slight negative skew

Kurtosis	3.588	Slightly heavier tails than a normal distribution
Durbin–Watson	2.001	No evidence of autocorrelation
Condition Number	3.02	No indication of serious multicollinearity



Significant Findings

The regression coefficients provide insights into how different factors influence students' Post-Semester GPA.

Positive Significant Predictors

- **Pre_Semester_GPA (β = 0.4607)** was the strongest positive predictor, indicating that students with higher prior academic performance are more likely to achieve higher semester-end GPAs.
- **Traditional_Study_Hours (β = 0.0708)** positively influenced academic performance, highlighting the continued importance of conventional study habits.
- **Year_of_Study (β = 0.0440)** showed a positive relationship with GPA, suggesting that students in higher academic years generally perform better.
- **Prompt_Engineering_Skill (β = 0.0230)** had a positive impact, indicating that students who interact more effectively with AI tools tend to perform better academically.
- **Weekly_GenAI_Hours (β = 0.0161)** also had a small but statistically significant positive effect, suggesting that moderate AI usage can support learning.
- Students using AI primarily for **Debugging/Troubleshooting (β = 0.0205)**
- **Primary_Use_Case_Debugging/Troubleshooting** intended to achieve slightly higher GPAs.

Negative Significant Predictors

- **Anxiety_Level_During_Exams (β = −0.0057)** showed a small negative relationship with GPA.
- **Primary_Use_Case_Direct_Answer_Generation (β = −0.0226)** negatively influenced GPA, suggesting that relying on AI to directly generate answers may reduce academic performance.
- The interaction term **Depending_Hours (β = −0.0194)**, representing the combined effect of Weekly GenAI Hours and Perceived AI Dependency, had a significant negative effect.

This indicates that although moderate AI usage may be beneficial, excessive AI usage combined with high dependency can negatively impact academic performance.

Interaction Effect

- **Perceived_AI_Dependency** was retained in the model to evaluate its interaction with **Weekly_GenAI_Hours**. While it is **not statistically**

- **significant on its own** ($p = 0.364$), the interaction term **Depending_Hours** is **significant and negative** ($\beta = -0.0194$, $p < 0.001$). This indicates that **higher AI usage hours combined with greater AI dependency is associated with lower Post-Semester GPA**, even though AI dependency alone does not significantly affect academic performance.

Summary Table for Strategic Action

Strategy Area	Insight from Model	Business Action
Curriculum	Prompting skill > Time spent	Mandate AI literacy training.
Product Design	Direct answers hurt performance	Design AI tutors to provide hints, not answers.
Risk Management	High dependency = High burnout	Flag high-dependency users for academic support.

Marketing/Messaging	Traditional study still drives GPA	Reinforce AI as a secondary aid to core study.

Overall Inference

The results indicate that **how students use Generative AI is more important than how much they use it**. Students who possess stronger Prompt Engineering skills and use AI primarily for learning-oriented activities, such as debugging and concept clarification, tend to achieve better academic performance than those who rely on AI for direct answer generation.

Traditional study habits remain a significant contributor to academic success, indicating that Generative AI should complement rather than replace independent learning. Furthermore, the interaction between AI usage and perceived AI dependency suggests that excessive reliance on AI may reduce the academic benefits associated with moderate AI usage.

From an institutional perspective, these findings highlight the importance of promoting responsible AI adoption through AI literacy programmes, assessment strategies that encourage critical thinking, and learning practices that balance AI-assisted and traditional study methods.

Model 2: Binary Logistic Regression

Objective

The objective of this model is to identify the academic, behavioural, AI-related, and psychological factors that significantly influence severe student burnout using Binary Logistic Regression and to provide interpretable insights for early identification and intervention.

Dependent Variable

Is_Burn_Out

- **Type:** Binary (Categorical)
- Coding:
    - **0** = Low/Medium Burnout Risk
    - **1** = High Burnout Risk

The binary target variable was created by combining the Low and Medium burnout categories into a single class while retaining the High Burnout category as the positive class. This transformation enables the model to focus specifically on distinguishing students experiencing severe burnout from those with relatively lower burnout risk.

Independent Variables

- Anxiety_Level_During_Exams
- Year_of_Study
- Tool_Diversity
- AI_Engagement
- AI_Dependency_Ratio
- Total_Study
- Perceived_AI_Dependency

- Skill_Retention_Score
- Pre_Semester_GPA

Feature Engineering

The engineered variables **AI_Engagement**, **AI_Dependency_Ratio**, **Total_Study**, and **Is_Burn_Out** were developed during the feature engineering stage described in **Section 3.4**. These variables were specifically designed to capture students' AI usage behaviour, study patterns, and severe burnout risk more effectively than the original dataset variables.

For this model, **Is_Burn_Out** was used as the dependent variable by combining the **Low** and **Medium** burnout categories into a single class while retaining **High Burnout** as the positive class. This transformation enabled the model to focus specifically on identifying students at risk of severe burnout.

Purpose:

The original burnout variable consisted of three categories (Low, Medium, and High). Exploratory analysis indicated substantial overlap between the Low and Medium categories. To focus specifically on identifying severe burnout, these categories were combined to create a binary outcome variable that distinguishes students experiencing high burnout from those with non-severe burnout.

Model Development

The original dependent variable, **Burnout_Risk_Level**, consisted of three categories: **Low**, **Medium**, and **High**. During exploratory analysis, considerable overlap was observed between the Low and Medium burnout groups, making it difficult to obtain a clear statistical distinction among all three categories. Since the primary objective of this study was to identify the factors associated with **severe burnout**, the outcome variable was transformed into a binary variable (**Is_Burn_Out**), where **High Burnout** was coded as **1** and the combined **Low and Medium Burnout** categories were coded as **0**. This transformation shifted the analytical focus from

differentiating minor variations in burnout to identifying students experiencing severe burnout who may require timely academic or psychological intervention.

Binary Logistic Regression was then employed because it is appropriate for modelling binary outcomes and estimating the relationship between multiple predictor variables and the likelihood of severe burnout. The model parameters were estimated using **Maximum Likelihood Estimation (MLE)** after standardising the predictor variables using **StandardScaler**. Although the dataset was divided into training and testing subsets during model development, the emphasis of this analysis is on interpreting the statistical relationships between the explanatory variables and severe burnout rather than evaluating predictive performance.

Model Assumptions & Limitations

The following assumptions were verified before estimating the Binary Logistic Regression model:

- The dependent variable is binary (Is_Burn_Out).
- Observations are independent, with each record representing a unique student.
- Predictor variables were standardised before model estimation.
- Multicollinearity was assessed using the Variance Inflation Factor (VIF). All predictor variables exhibited VIF values below **3**, indicating no evidence of problematic multicollinearity.
- A sufficiently large sample size (29,999 observations for model estimation) ensured stable coefficient estimation using Maximum Likelihood Estimation.

- Assumes a linear relationship between predictor variables and the log-odds of severe burnout.
- Binary Logistic Regression may not fully capture complex non-linear interactions among predictors.
- The predictive performance depends on the quality and representativeness of the available dataset.
- The model is sensitive to class imbalance, which may influence prediction of the minority class.

- External validation is recommended before generalising the model to other educational settings.

Model Evaluation

The Binary Logistic Regression model was evaluated from both **inferential** and **predictive** perspectives. While inferential evaluation assesses the statistical significance and explanatory capability of the model, predictive evaluation examines its effectiveness in correctly identifying students at high risk of severe burnout. Together, these evaluation measures provide a comprehensive assessment of the model's reliability and practical applicability.

A. Statistical Model Evaluation

Metric	**Value**
Estimation Method	Maximum Likelihood Estimation (MLE)
Number of Observations	29,999
Model	Binary Logistic Regression (Logit)
Pseudo R^2 (McFadden)	**0.2013**
Log-Likelihood	**−13,467**
LL-Null	**−16,861**

LLR p-value	**< 0.001**
Convergence	**Successful (6 Iterations)**

Interpretation

- The model converged successfully after **6 iterations**, indicating stable coefficient estimation using Maximum Likelihood Estimation (MLE).
- The **Likelihood Ratio Test (LLR p < 0.001)** confirms that the model is statistically significant and performs better than the intercept-only model.
- The **Pseudo R^2 of 0.2013** suggests that the selected predictors provide meaningful explanatory power for identifying students at high burnout risk.
- The improvement in **Log-Likelihood** from **−16,861** to **−13,467** indicates that the inclusion of academic, behavioural, AI-related, and psychological variables significantly enhances the model's predictive capability.

Overall, the results demonstrate that the Binary Logistic Regression model is statistically reliable for identifying factors associated with severe student burnout.

B. Classification Performance

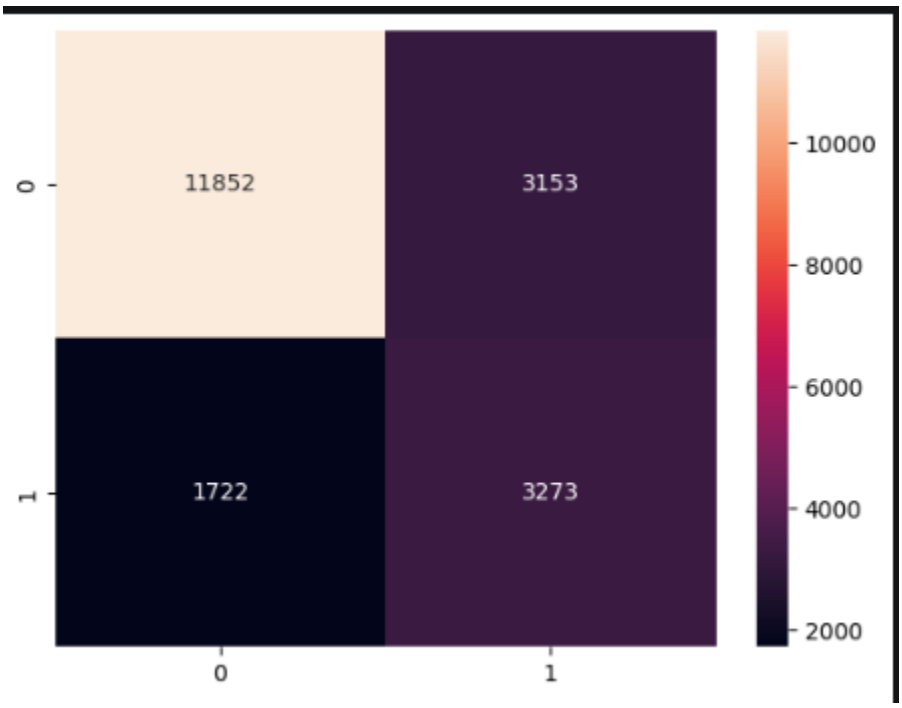
Metric	**Value**
Accuracy	**75.63%**

Precision (High Burnout)	**51%**
Recall (High Burnout)	**66%**
F1-Score (High Burnout)	**57%**
ROC-AUC	**0.793**
Classification Threshold	**0.2579**

Interpretation

The Binary Logistic Regression model achieved an **accuracy of 75.6%**, indicating satisfactory classification performance. Using an optimal threshold of **0.2579** (determined through Youden's Index), the model improved its ability to identify students at high burnout risk. It attained a **recall of 66%**, successfully identifying nearly two-thirds of severely burned-out students, while a **precision of 51%** indicates that about half of the predicted high-risk cases were correctly classified. The **F1-score of 57%** reflects a reasonable balance between precision and recall, supporting the model's effectiveness as an early screening tool.

C. Confusion Matrix Analysis



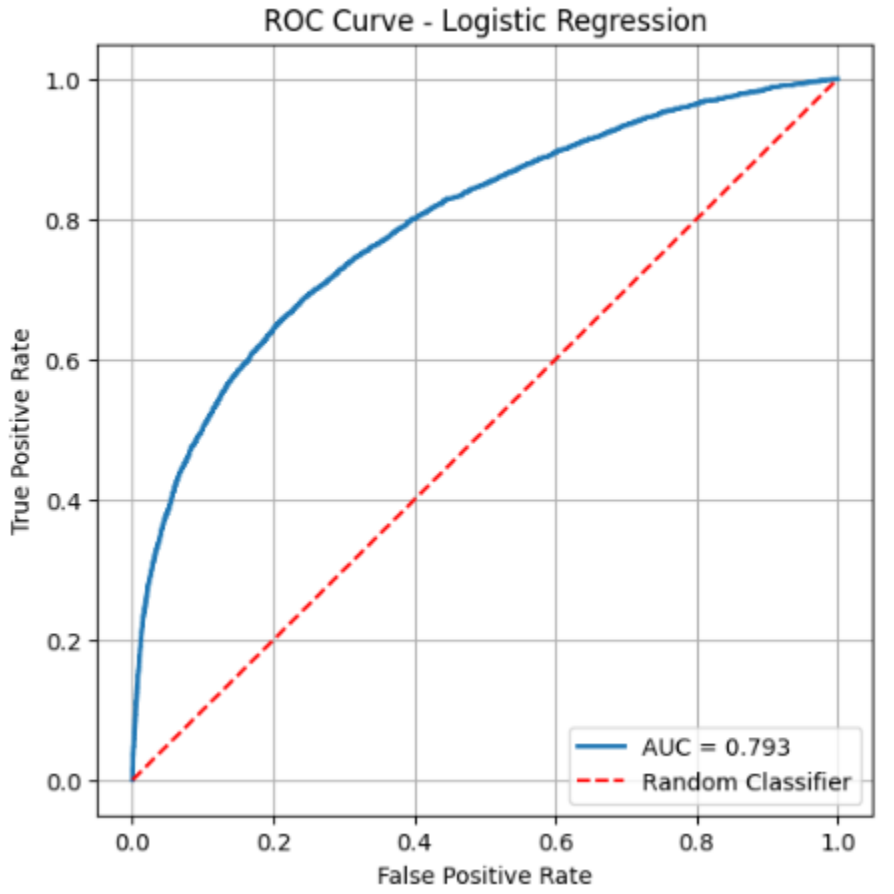
Interpretation

The model correctly identified **11,852** students belonging to the Low/Medium Burnout category and **3,273** students experiencing High Burnout. However, **1,722** High Burnout students were incorrectly classified as Low/Medium (False Negatives), while **3,153** Low/Medium students were incorrectly identified as High Burnout (False Positives).

Considering the objective of this study, the current balance between False Positives and False Negatives is acceptable, as failing to identify genuinely high-risk students may delay timely academic or psychological intervention, whereas False Positives would primarily result in additional monitoring or counselling. Furthermore, some False Positive cases may represent **borderline students** whose characteristics are more similar to the High Burnout group despite being labelled as Low or Medium. Given the overlap observed between the original burnout categories, these cases may reflect ambiguity in class boundaries rather than model error alone,

further supporting the decision to distinguish **severe burnout** from **non-severe burnout** through binary classification.

D. ROC Curve Analysis



The ROC Curve assessed the model's ability to distinguish between students with and without severe burnout across different classification thresholds. The model achieved an **AUC of 0.793**, indicating good discriminative ability. Based on the ROC analysis, an optimal threshold of **0.2579** was selected, resulting in a **True Positive Rate (Recall) of 65.53%** and a **False Positive Rate of 21.01%**. This threshold provides a balanced trade-off between identifying high-risk students and minimizing false alarms, making the model suitable for early burnout screening in educational settings.

Significant Findings

The regression coefficients reveal several statistically significant predictors influencing severe student burnout.

Positive Significant Predictors

- **Pre_Semester_GPA (β = 0.226, p < 0.001)** indicates that students with stronger prior academic performance were more likely to experience severe burnout after controlling for other variables. This may reflect greater academic expectations and workload among high-achieving students.
- **AI_Dependency_Ratio (β = 0.4495)** emerged as one of the strongest predictors of severe burnout, suggesting that students whose AI usage substantially exceeds their traditional study time are significantly more likely to experience burnout.
- **AI_Engagement (β = 0.3280)** was also positively associated with burnout, indicating that intensive engagement with multiple AI tools may contribute to increased academic pressure and cognitive overload.
- **Perceived_AI_Dependency (β = 0.3534)** significantly increased the likelihood of severe burnout, highlighting that psychological reliance on AI extends beyond simple usage frequency.
- **Year_of_Study (β = 0.3767)** showed that students in higher academic years experience greater burnout risk, likely due to increasing academic responsibilities.
- **Anxiety_Level_During_Exams (β = 0.0973)** exhibited a positive relationship with severe burnout, indicating that examination-related stress remains an important psychological determinant.
- **Total_Study (β = 0.3403)** was positively associated with burnout, suggesting that increased overall study workload contributes to elevated burnout risk.

Negative Significant Predictors

- **Tool_Diversity (β = −0.1440)** reduced the likelihood of severe burnout, suggesting that students using a broader range of AI tools may utilise technology more efficiently instead of relying excessively on a single platform.
- **Skill_Retention_Score (β = −0.0766)** demonstrated a protective effect, indicating that students who retain knowledge more effectively are less likely to experience severe burnout.

Overall Inference

The findings indicate that severe student burnout is influenced by a combination of academic workload, AI usage behaviour, psychological dependency, and learning characteristics rather than any single factor. Among the predictors, **AI Dependency Ratio**, **Perceived AI Dependency**, **Year of Study**, and **AI Engagement** exhibited the strongest positive associations with severe burnout, suggesting that excessive reliance on AI, particularly when accompanied by increasing academic demands, significantly elevates burnout risk. Conversely, greater **Tool Diversity** and **Skill Retention Score** were associated with a lower likelihood of severe burnout, indicating that effective learning strategies and diversified AI usage may act as protective factors.

Severe burnout can also be seen through students who have greater AI dependency. It might be because of other factors but AI dependency and usage is surely a part of it. A greater diversity of AI tools may enable students to complete academic tasks more efficiently, thereby reducing workload and lowering the likelihood of severe burnout.

Business Recommendations

1. Integrate Prompt Engineering and AI-Assisted Problem Solving into the Curriculum

The study indicates that **Prompt Engineering Skill contributes positively to academic performance**, while students using AI primarily for debugging and concept clarification achieve better academic outcomes than those relying on direct answer generation. Universities should therefore formally integrate Prompt Engineering, AI-assisted problem solving, and critical evaluation of AI-generated responses into the curriculum. This would enable students to leverage AI more effectively while strengthening analytical and problem-solving skills rather than encouraging passive dependence on AI-generated content.

2. Redesign Assessment Methods to Promote Higher-Order Thinking

The findings suggest that students relying on AI for **Direct Answer Generation** perform worse academically than those using AI for learning-oriented tasks such as debugging and troubleshooting. This indicates that conventional assignments are increasingly susceptible to AI-generated solutions. Universities should redesign assessment frameworks by incorporating case studies, project-based learning, presentations, practical problem-solving exercises, and application-oriented questions that evaluate reasoning, creativity, and decision-making rather than simple answer reproduction. Such assessments would encourage meaningful AI usage while preserving academic integrity.

3. Develop an AI Learning Analytics Dashboard for Early Academic Intervention

The Linear Regression and Binary Logistic Regression models together demonstrate that academic performance and severe burnout can be explained using behavioural, academic, and AI-related variables. Universities should therefore implement an AI learning analytics dashboard that monitors indicators such as AI Dependency Ratio, Weekly AI Usage, Traditional Study Hours,

Prompt Engineering Skill, Anxiety Level, and Burnout Risk Probability. Rather than serving as a surveillance mechanism, this dashboard should function as a decision-support tool for academic advisors, enabling timely intervention for students exhibiting unhealthy learning patterns before these adversely affect academic performance or well-being.

4. Establish AI Literacy and Tool Diversification Programmes

The study identifies **Tool Diversity** as a protective factor against severe burnout while demonstrating that effective AI engagement is more beneficial than excessive dependence on a single AI platform. Universities should organise periodic AI literacy programmes, workshops, hackathons, and tool demonstration sessions to expose students to a diverse range of AI applications and their appropriate use cases. Such initiatives can help students build efficient AI workflows, improve digital adaptability, and reduce excessive reliance on any single AI tool.

5. Introduce Progressive AI Support Across Academic Years

The analysis indicates that the likelihood of severe burnout increases with **Year of Study**, suggesting that students experience different academic pressures as they progress through university. Rather than providing identical AI support throughout the programme, universities should adopt a staged AI integration strategy. Early academic years should focus on AI literacy, Prompt Engineering, and responsible AI usage, while senior years should emphasise research assistance, project management, advanced AI applications, and burnout management strategies aligned with increasing academic demands.

Conclusion

This study examined the impact of Generative Artificial Intelligence (GenAI) on university students from two complementary perspectives: **academic performance** and **student well-being**. Using Multiple Linear Regression and Binary Logistic Regression, the study investigated how academic, behavioural, AI-related, and psychological factors influence students' Post-Semester GPA and the likelihood of severe burnout. The study provides a comprehensive understanding of the opportunities and challenges associated with AI adoption in higher education.

The findings demonstrate that **the quality of AI usage is more important than the quantity of AI usage**. Students with stronger Prompt Engineering skills and those who primarily use AI for learning-oriented tasks such as debugging and concept clarification tend to achieve better academic performance. In contrast, excessive AI dependency, intensive AI engagement, increasing academic workload, and examination anxiety are associated with a higher likelihood of severe burnout. The study also highlights that traditional study habits continue to play a significant role in academic success, indicating that AI should complement rather than replace independent learning.

From an institutional perspective, the results suggest that universities should move beyond simply encouraging or restricting AI adoption and instead focus on developing students' ability to use AI effectively and responsibly. Curriculum design, assessment strategies, learning analytics, and student support systems should be adapted to ensure that AI enhances learning while minimizing the risks associated with dependency and burnout.

Overall, this study demonstrates that the successful integration of Generative AI in higher education requires balancing academic performance with student well-being. The findings provide evidence-based insights that can help educational institutions promote responsible AI usage, enhance learning outcomes, and create supportive AI-enabled learning environments.

References

- Kaggle. (2025). *AI Impact on Students* [Dataset]. Retrieved from https://www.kaggle.com/datasets/laveshjadon/ai-impact-on-students.

- *statsmodels Development Team.* (2025). *statsmodels: Statistical modeling in Python* (Version 3.12). Retrieved from https://www.statsmodels.org.

Appendix : Sample Screenshots

Appendix 1 - Dataset Preview

- First few rows of the dataset (df.head())

	Student_ID	Major_Category	Year_of_Study	Pre_Semester_GPA	Weekly_GenAI_Hours	Primary_Use_Case	Prompt_Engineering_Skill	Tool_Diversity	Paid_Subscription	Traditional_Study_Hours	Perceived
35169	135170	Medical	Freshman	2.258	10.04	Debugging/Troubleshooting	Intermediate	3	True	7.77	
44394	144395	STEM	Freshman	3.401	10.47	Debugging/Troubleshooting	Intermediate	2	False	14.05	
45549	145550	Humanities	Senior	3.122	3.59	Copywriting/Drafting	Intermediate	5	False	16.28	
11059	111060	Business	Senior	3.058	3.02	Ideation	Beginner	1	False	4.67	
36125	136126	Medical	Junior	3.379	2.48	Ideation	Advanced	4	True	9.07	
38431	138432	Business	Graduate	3.523	1.06	Ideation	Advanced	5	True	12.47	
13434	113435	Humanities	Freshman	1.881	1.25	Copywriting/Drafting	Intermediate	4	False	8.12	
40010	140011	Medical	Sophomore	3.690	2.20	Direct_Answer_Generation	Intermediate	2	True	9.73	
16841	116842	Arts	Junior	2.879	8.87	Direct_Answer_Generation	Intermediate	3	False	13.19	

Appendix 2 - Data Preprocessing

- Data information (df.info())

```
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 16 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Student_ID                 50000 non-null  int64
 1   Major_Category             50000 non-null  object
 2   Year_of_Study              50000 non-null  object
 3   Pre_Semester_GPA           50000 non-null  float64
 4   Weekly_GenAI_Hours         50000 non-null  float64
 5   Primary_Use_Case           50000 non-null  object
 6   Prompt_Engineering_Skill   50000 non-null  object
 7   Tool_Diversity             50000 non-null  int64
 8   Paid_Subscription          50000 non-null  bool
 9   Traditional_Study_Hours    50000 non-null  float64
 10  Perceived_AI_Dependency    50000 non-null  int64
 11  Institutional_Policy       50000 non-null  object
 12  Anxiety_Level_During_Exams 50000 non-null  int64
 13  Post_Semester_GPA          50000 non-null  float64
 14  Skill_Retention_Score      50000 non-null  float64
 15  Burnout_Risk_Level         50000 non-null  object
dtypes: bool(1), float64(5), int64(4), object(6)
```

- Feature engineering output

```
train_data["Depending_Hours"] = (train_data.Weekly_GenAI_Hours - train_data.Weekly_GenAI_Hours.mean()) * (train_data.Perceived_AI_Dependency - train_data.Perceived_AI_Dependency.mean())
```

```
copy2["AI_Engagement"] = (
    copy2["Weekly_GenAI_Hours"] *
    copy2["Tool_Diversity"] *
    copy2["Prompt_Engineering_Skill"])

copy2["AI_Dependency_Ratio"] = (
    copy2["Weekly_GenAI_Hours"] /
    (copy2["Traditional_Study_Hours"] + 1)
)

copy2["Total_Study"] = (
    copy2["Weekly_GenAI_Hours"] +
    copy2["Traditional_Study_Hours"]
)

copy2["GPA_Change"] = (
    copy2["Post_Semester_GPA"] -
    copy2["Pre_Semester_GPA"]
)
```

Appendix 3 - Model Development

- Linear Regression summary output

OLS Regression Results

Dep. Variable:	Post_Semester_GPA	R-squared:	0.895
Model:	OLS	Adj. R-squared:	0.895
Method:	Least Squares	F-statistic:	2.615e+04
Date:	Tue, 07 Jul 2026	Prob (F-statistic):	0.00
Time:	18:17:05	Log-Likelihood:	12226.
No. Observations:	29999	AIC:	-2.443e+04
Df Residuals:	29988	BIC:	-2.434e+04
Df Model:	10		
Covariance Type:	HC3		

	coef	std err	z	P>|z|	[0.025	0.975]
const	3.3493	0.001	3599.665	0.000	3.347	3.351
Year_of_Study	0.0440	0.001	47.383	0.000	0.042	0.046
Pre_Semester_GPA	0.4607	0.001	501.430	0.000	0.459	0.462
Weekly_GenAI_Hours	0.0161	0.002	9.943	0.000	0.013	0.019
Prompt_Engineering_Skill	0.0230	0.001	24.308	0.000	0.021	0.025
Traditional_Study_Hours	0.0708	0.001	74.504	0.000	0.069	0.073
Perceived_AI_Dependency	-0.0011	0.001	-0.908	0.364	-0.004	0.001
Anxiety_Level_During_Exams	-0.0057	0.001	-5.696	0.000	-0.008	-0.004
Primary_Use_Case_Debugging/Troubleshooting	0.0205	0.001	21.496	0.000	0.019	0.022
Primary_Use_Case_Direct_Answer_Generation	-0.0226	0.001	-19.492	0.000	-0.025	-0.020
Depending_Hours	-0.0194	0.002	-11.533	0.000	-0.023	-0.016

Omnibus:	428.573	Durbin-Watson:	2.001
Prob(Omnibus):	0.000	Jarque-Bera (JB):	599.299
Skew:	-0.183	Prob(JB):	7.31e-131
Kurtosis:	3.588	Cond. No.	3.02

- Logistic Regression summary output

```
Optimization terminated successfully.
         Current function value: 0.448931
         Iterations 6
```

Logit Regression Results

Dep. Variable:	Is_Burn_Out	No. Observations:	29999
Model:	Logit	Df Residuals:	29989
Method:	MLE	Df Model:	9
Date:	Fri, 10 Jul 2026	Pseudo R-squ.:	0.2013
Time:	23:45:56	Log-Likelihood:	-13467.
converged:	True	LL-Null:	-16861.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P>|z|	[0.025	0.975]
const	-1.3027	0.016	-81.049	0.000	-1.334	-1.271
Anxiety_Level_During_Exams	0.0973	0.016	6.202	0.000	0.067	0.128
Year_of_Study	0.3767	0.016	24.170	0.000	0.346	0.407
Tool_Diversity	-0.1440	0.021	-6.984	0.000	-0.184	-0.104
AI_Engagement	0.3280	0.027	12.151	0.000	0.275	0.381
AI_Dependency_Ratio	0.4495	0.023	19.919	0.000	0.405	0.494
Total_Study	0.3403	0.021	16.128	0.000	0.299	0.382
Perceived_AI_Dependency	0.3534	0.020	17.723	0.000	0.314	0.393
Skill_Retention_Score	-0.0766	0.016	-4.774	0.000	-0.108	-0.045
Pre_Semester_GPA	-0.2267	0.015	-15.004	0.000	-0.256	-0.197